AP PRECALCULUS

Unit 1 • Lesson 1.4 notes

Lesson 1.4 • Polynomial Functions and Rates of Change

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Identify the degree, leading term and leading coefficient of a polynomial.
- Locate local and global extrema from a graph.
- Identify points of inflection as changes in concavity.

Vocabulary. polynomial, degree, leading coefficient, local maximum, local minimum, global (absolute) extremum, point of inflection.

1. Polynomial functions

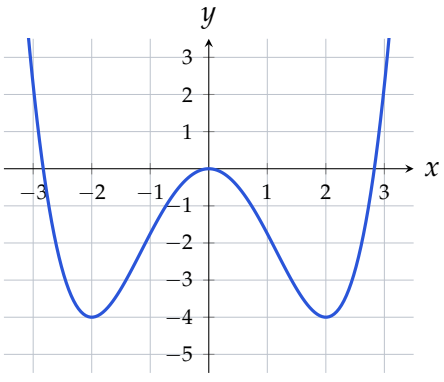
$p(x) = a_n x^n + \cdots + a_1 x + a_0$ with $a_n \neq 0$ has degree n. To find the degree of a product, add the degrees of its factors: $(x-1)^2(x+4)$ has degree 3.

2. Extrema

Key idea

- **Local maximum** at $x = a$: p changes from increasing to decreasing; $p(a)$ is the greatest value nearby.
- **Local minimum** at $x = a$: p changes from decreasing to increasing.
- The **global maximum** on an interval is the greatest value of p on that interval (it may be at an endpoint).
- A polynomial of degree n has at most $n-1$ local extrema and its graph has no breaks.

Example 1 • Reading extrema



The graph of $p(x) = \frac{1}{4}x^4 - 2x^2$ is shown.

Solution. Local minima at $(-2,-4)$ and $(2,-4)$, local maximum at $(0,0)$. The global minimum value is -4; there is no global maximum on $(-\infty,\infty)$. On $[0,3]$, the global maximum is $p(3) = \frac{81}{4} - 18 = 2.25$, at the endpoint.

3. Points of inflection

A point of inflection is where the graph changes concavity, that is, where the rate of change switches from increasing to decreasing or the reverse. Between a local maximum and the next local minimum, a polynomial must have at least one point of inflection.

Example 2

$p(x) = x^3 - 6x^2 + 9x$ has a local maximum at $(1, 4)$, a local minimum at $(3, 0)$ and a point of inflection at $(2, 2)$.
Solution. p is increasing on $(-\infty, 1)$, decreasing on $(1, 3)$, increasing on $(3, \infty)$. The rate of change is decreasing until $x = 2$ (concave down) and increasing after $x = 2$ (concave up).

Common mistakes

- Giving the x-value when the question asks for the extreme *value* (the y-value), or the reverse.
- Forgetting to check the endpoints when looking for a global extremum on a closed interval.

Quick check

1. What is the maximum number of turning points of a degree-6 polynomial?
2. Can a cubic polynomial have no local extrema? Give an example.

Answers. 1. 5. 2. Yes, for example $p(x) = x^3$, which is always increasing.